

# Professional Vue.js

Version 2.0, 2026

In this 3-day course, students will learn the fundamental concepts and structures of Vue.js 3 development, along with the modern JavaScript tooling that supports it: Node.js, npm, Vite, ESLint, and Vitest. Students will use Vue.js, the Composition API, Pinia, and Vue Router to build a real-world frontend application through a series of 24 hands-on labs. An optional review of modern JavaScript is included for students who need it.

## TARGET AUDIENCE

This course is designed for experienced front-end web developers who wish to begin using Vue.js.

## OBJECTIVES

After taking this course, students will be able to:

- Install, configure, and use modern web tooling, including Node.js, npm, Vite, and ESLint
- Understand what Vue.js is and what problem it solves
- Explore the basic architecture of a Vue.js application
- Gain a deep knowledge of Vue.js components and the Composition API
- Reuse stateful logic with composables
- Use Pinia for maintaining state in a Vue.js application
- Build a Single Page Application with Vue Router
- Test Vue components and stores with Vitest and Vue Test Utils
- Debug Vue applications with the browser developer tools and Vue DevTools
- Learn Vue.js best practices

## TOPICS

- Node.js and npm
- Modern JavaScript tooling: Vite, ESLint, and transpilers
- The Document Object Model (DOM)
- Debugging JavaScript and Vue applications
- The Composition API and <script setup>
- Vue template syntax and directives
- Reactivity, data binding, and computed properties
- Event handling, watchers, and the component lifecycle
- Forms
- Communication between components with props, events, and slots
- Composables
- State management with Pinia
- Communication with the server via Ajax
- Creating SPAs with Vue Router
- Unit testing with Vitest and Vue Test Utils
- Animation and Transitions in Vue (time permitting)
- Modern JavaScript review, including promises and async/await (optional)

## Prerequisites

Before taking this course, students should have solid experience with HTML5, CSS3, and JavaScript

## Duration

3 days

## OUTLINE

1. Introduction and Vue Quickstart
  - a. What is Vue.js?
  - b. What is Reactivity?
  - c. History of Vue.js
  - d. Vue vs. React and Angular
  - e. Virtual DOM
  - f. What Vue 3 Changed
  - g. Code Editors and IDEs
  - h. Lab 01: Installing and Configuring VS Code
  - i. Vite and create-vue
  - j. Lab 02: Get Started with Vite
  - k. Installing and Using the Vue DevTools
  - l. Our Real-world Project: Conduit
  - m. Single-File Components
  - n. Lab 03: Your First Component
  - o. Lab 04: Create More Components
  - p. Lab 05: Testing Vue
2. Node.js
  - a. What is Node.js?
  - b. The Event Loop
  - c. EventEmitter
  - d. Modules
  - e. Front-end Node
  - f. Lab 06: Getting Started with Node.js and npm
3. Tooling
  - a. Why Have Tooling?
  - b. npm and Other Package Managers
  - c. Lab 07: Initialize npm
  - d. node\_modules and package.json
  - e. Lab 08: Making a Web Server
  - f. Static Code Analysis with ESLint
  - g. Lab 09: Automate Linting
  - h. Transpilers: Babel and esbuild
  - i. Bundlers and How Vite Works
4. The DOM and Debugging
  - a. What is the DOM?
  - b. Nodes, EventTarget, and DOM Events
  - c. Manipulating HTML with the DOM, jQuery, React, and Vue
  - d. Browser Development Tools
  - e. Vue DevTools

- f. Lab 10: Manual In-Browser Testing and Debugging
- 5. Basic Vue.js Features
  - a. Options API vs. Composition API
  - b. Creating and Mounting an App
  - c. `<script setup>`, `ref`, and `computed`
  - d. Compiler Macros
  - e. App-level Configuration
  - f. Vue Template Syntax
  - g. Directives, Arguments, and Modifiers
  - h. Custom Directives
  - i. Loops and Lists with `v-for` and `key`
  - j. Conditional Rendering: `v-if` vs. `v-show`
  - k. Lab 11: Static Version
  - l. Binding HTML Classes and Styles
  - m. Outputting Raw HTML
  - n. Lab 12: Styling Vue Components
  - o. Computed Properties
  - p. Filtering and Sorting with Computed Properties
  - q. Lab 13: Computed Properties
- 6. Reactivity, Events, and Forms
  - a. Reactive State with `ref` and `reactive`
  - b. Data Binding with `v-bind` and `v-model`
  - c. Reactivity Caveats
  - d. Lab 14: Methods and State
  - e. Event Handling and Event Modifiers
  - f. Emitting Events from Child Components
  - g. Lab 15: Events
  - h. Watchers: `watch` and `watchEffect`
  - i. Component Lifecycle Hooks
  - j. Lab 16: Component Lifecycle
  - k. Input Bindings
  - l. Checkboxes, Radio Buttons, and Select Elements
  - m. Input Modifiers
  - n. Lab 17: Forms
- 7. Vue Components
  - a. Creating, Organizing, and Naming Components
  - b. Local vs. Global Components
  - c. Props
  - d. Fragments: Multiple Root Elements
  - e. `v-model` on Components and `defineModel`
  - f. Content Distribution with Slots
  - g. Named Slots
  - h. Lab 18: Slots
  - i. Loading Components Asynchronously

8. Composables
  - a. Composable Basics
  - b. Why Composables Replace Mixins
  - c. Writing and Using Composables
  - d. Composable Conventions
  - e. Community Composables
  - f. Lab 19: Composables
9. State Management with Pinia
  - a. Why a Store?
  - b. Pinia vs. Vuex
  - c. Defining a Store and Setup Stores
  - d. State, Getters, and Actions
  - e. Using a Store in a Component
  - f. storeToRefs
  - g. Multiple Stores
  - h. Pinia and the DevTools
  - i. Persisting Store State
  - j. Testing a Store
  - k. Lab 20: Implementing Pinia
10. Single-Page Applications
  - a. Creating an SPA with Vue Router
  - b. Lab 21: Routing
  - c. Lab 22: AJAX
11. Testing
  - a. Test-Driven Development
  - b. Vitest
  - c. Test Suites, Specs, and Matchers
  - d. TDD vs. BDD
  - e. Mocking and Fake Timers
  - f. Snapshot Testing
  - g. Vue Test Utils: mount and shallowMount
  - h. Lab 23: Testing with Vitest
  - i. Browser and End-to-End Testing
12. Transitions and Animations
  - a. <Transition>
  - b. Transition Classes
  - c. JavaScript Hooks
  - d. Custom Transition Classes
  - e. Lab 24: Transitions and Animations
13. JavaScript Review (Optional)
  - a. Variable Scoping with const and let
  - b. Arrow Functions
  - c. Default Parameter Handling
  - d. Rest and Spread

- e. Template Literals
- f. Enhanced Object Properties
- g. Method Notation
- h. Array Matching
- i. Object Matching
- j. Symbol Primitive
- k. User-defined Iterators
- l. For-Of Operator
- m. Creating and Consuming Generator Functions
- n. Classes
- o. Beyond ES2015
- p. Strict Mode
- q. Understanding this
- r. Array.map()
- s. Promises
- t. Async / Await